GEOGRAPHY 03 — VULCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES

ENDOGENIC SYSTEM: PLATE MOTION → EARTHQUAKE SOURCE GEOMETRY, DEPTH → SEISMIC WAVES, SHADOW ZONES → GLOBAL SEISMIC BELTS, CASCADING → VOLCANO ANATOMY, MAGMA CONTROLS → VOLCANIC SETTINGS, GLOBAL BELTS

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g6 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 ENDOGENIC SYSTEM: PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION

ENDOGENIC SYSTEM | PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION | INTERNAL HEAT + GRAVITY + PLATE MOTION RESHAPE CRUST. | STRESS ACCUMULATION MAGMA GENERATION LOAD | STRAIN DIVERGENCE | DECOMPRESSION SUBDUCTION | VOLATILE FLUX RUPTURE | SLIP PLUME

ENDOGENIC SYSTEM → Internal heat + gravity + plate motion reshape crust. → STRESS ACCUMULATION MAGMA GENERATION load → lock → strain divergence: decompression subduction: volatile flux rupture → anomalous upwelling → EARTHQUAKE VULCANISM landforms + hazards + resources

DEFINITION / COMPONENTS
- ENDOGENIC SYSTEM
- Internal heat + gravity + plate motion reshape crust.
- STRESS ACCUMULATION MAGMA GENERATION load → lock → strain divergence: decompression subduction: volatile flux rupture → slip plume

TRIGGER / MECHANISM
- anomalous upwelling
- EARTHQUAKE VULCANISM landforms + hazards + resources
- FAULT PAIRS tension → normal

EFFECT / INTERVENTION
- compression → reverse/thrust
- shear → strike-slip
- Plate motion loads a region; sudden fault rupture is the immediate quake source.

MECHANISM / VERDICT — Plate tectonics gives the spatial grammar of seismicity and vulcanism, but the examiner still expects the immediate mechanics—fault slip for an earthquake and magma ascent for an eruption.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate tectonics gives the spatial grammar of seismicity and vulcanism, but the examiner still expects the immediate mechanics—fault slip for an earthquake and magma ascent for an eruption.

01

STAGE 01 EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC

EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC | SURFACE STATION A -------- EPICENTRE -------- STATION B FOCUS | HYPOCENTRE RUPTURE BEGINS ON FAULT PLANE | SEISMOGRAM ARRIVAL ORDER | P FIRST | S SECOND | LOVE AND RAYLEIGH LATER | P-S INTERVAL AT ONE STATION GIVES SOURCE DISTANCE THREE

SURFACE station A -------- epicentre -------- station B focus / hypocentre rupture begins on fault

SEISMOGRAM ARRIVAL ORDER → P first → S second → Love and Rayleigh later → P-S interval at one station gives source distance three or more distance circles estimate the

SYSTEM / DEFINITION
- SURFACE station A -------- epicentre -------- station B focus / hypocentre rupture begins on fault plane
- SEISMOGRAM ARRIVAL ORDER
- P first → S second → Love and Rayleigh later

PROCESS / MECHANISM
- P-S interval at one station gives source distance three or more distance circles estimate the epicentre
- DEPTH CLASS shallow: 0-70 km
- intermediate: 70-300 km

SPATIAL / INDIA PATTERN
- deep: 300-700 km
- SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels
- a swarm has no single clearly dominant event.

HAZARD / LIMIT / RESPONSE
- Epicentre is a vertical projection, not necessarily the worst-damaged place.

MECHANISM / VERDICT — SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels;.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels;.

02

STAGE 02 SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY | MAIN CLUE | SOLID, LIQUID | CORE REFRACTION | TRANSVERSE SHEAR | SOLID ONLY | LIQUID OUTER CORE | HORIZONTAL SHEAR

WAVE	MOTION	MEDIUM	MAIN CLUE
P	compression	solid, liquid	core refraction
S	transverse shear	solid only	liquid outer core
Love	horizontal shear	surface	lateral shaking
Rayleigh	rolling ellipse	surface	mixed motion
P shadow about 103-142 deg	direct S shadow beyond about 103 deg		

DEFINITION / COMPONENTS
- MAIN CLUE
- solid, liquid

TRIGGER / MECHANISM
- liquid outer core
- horizontal shear

EFFECT / INTERVENTION
- P shadow about 103-142 deg
- direct S shadow beyond about 103 deg

MECHANISM / VERDICT — SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels;.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels;.

02

STAGE 02 SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY · MAIN CLUE · SOLID, LIQUID · CORE REFRACTION · TRANSVERSE SHEAR · SOLID ONLY · LIQUID OUTER CORE · HORIZONTAL SHEAR

WAVE	MOTION	MEDIUM	MAIN CLUE
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DEFINITION / COMPONENTS
- MAIN CLUE
- solid, liquid
- core refraction
- transverse shear
- solid only

TRIGGER / MECHANISM
- liquid outer core
- horizontal shear
- lateral shaking
- rolling ellipse
- mixed motion

EFFECT / INTERVENTION
- P shadow about 103-142 deg
- direct S shadow beyond about 103 deg
- MAGNITUDE: one source measure; Mw uses seismic moment moment = rigidity x fault area x average slip
- 1 local magnitude gives about 10x amplitude and 31.6x energy
- INTENSITY: place-specific observed effects; many values for one event source x path x site x exposure x vulnerability / capacity → damage

MECHANISM / VERDICT — Focal depth is a source descriptor, whereas disaster severity is a relational outcome produced by source, path, site, exposure and vulnerability.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Focal depth is a source descriptor, whereas disaster severity is a relational outcome produced by source, path, site, exposure and vulnerability.

03

STAGE 03 GLOBAL SEISMIC BELTS, CASCADING HAZARDS AND TSUNAMI SEQUENCE

GLOBAL SEISMIC BELTS, CASCADING HAZARDS AND TSUNAMI SEQUENCE · GLOBAL EARTHQUAKE DISTRIBUTION · RIDGES/TRANSFORMS TRENCH + SLAB + ARC · HIMALAYAN COLLISION · SHALLOW EVENTS MEGATHRUST + TSUNAMI · COMPLEX CONVERGENCE · BASALTIC RIDGES · INTRAPLATE EVENTS REACTIVATE INHERITED STRUCTURES; STABLE IS NOT ASEISMIC.

GLOBAL EARTHQUAKE DISTRIBUTION → Ridges/transforms trench + slab + arc → Himalayan collision → shallow events megathrust + tsunami → complex convergence

DEFINITION / COMPONENTS
- GLOBAL EARTHQUAKE DISTRIBUTION
- Ridges/transforms trench + slab + arc
- Himalayan collision
- shallow events megathrust + tsunami
- complex convergence

TRIGGER / MECHANISM
- basaltic ridges
- Intraplate events reactivate inherited structures; stable is not aseismic.
- CASCADE shaking → amplification / liquefaction / lateral spreading
- landslide / fire / lifeline failure / dam distress
- TSUNAMI submarine megathrust → vertical sea-floor displacement

EFFECT / INTERVENTION
- long low deep-ocean wave → coastal shoaling
- run-up + inundation + return flow
- First wave need not be largest; withdrawal need not occur before every event.
- Risk varies with bathymetry, bay shape, shelf, exposure and evacuation.

MECHANISM / VERDICT — Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

04

STAGE 04 VOLCANO ANATOMY, MAGMA CONTROLS, LANDFORMS AND ERUPTION STYLES

VOLCANO ANATOMY, MAGMA CONTROLS, LANDFORMS AND ERUPTION STYLES · ASH PLUME · \ CRATER LAVA -- -- LAVA VENT DYKE · SILL CONDUIT MAGMA CHAMBER · MAGMA BELOW SURFACE · LAVA AFTER ERUPTION · SILICA UP · VISCOSITY UP

ash plume

\ crater lava -- -- lava vent dyke → sill conduit magma chamber → Magma below surface

CLASSIFICATION
- ash plume
- \ crater lava -- -- lava vent dyke
- sill conduit magma chamber
- Magma below surface

DIAGNOSTIC BASIS
- lava after eruption
- Silica up → viscosity up → degassing harder → explosivity may rise
- Temperature up → viscosity generally down
- Retained gas + decompression → bubbles → fragmentation

PROCESS LINK
- FORMS: batholith, laccolith, lopolith, phacolith, dyke, sill
- Dyke cuts layers; sill follows layers.
- STYLE: fissure/Icelandic → Hawaiian → Strombolian → Vulcanian → Plinian
- FORM: shield

EXAM TRAP
- cinder cone
- fissure plateau
- STATUS: active
- extinct; status is not a compulsory life cycle.

MECHANISM / VERDICT — Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

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ash plume

\ crater lava -- -- lava vent dyke → sill conduit magma chamber → Magma below surface

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EXAM TRAP
- cinder cone
- fissure plateau
- STATUS: active
- extinct; status is not a compulsory life cycle.

MECHANISM / VERDICT — Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

05

STAGE 05 VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS

VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS · MELTING MECHANISM · RIDGE, RIFT, BASALT · SLAB VOLATILE FLUX · TRENCH-SLAB-ARC SYSTEM · MANTLE UPWELLING · FLOOD BASALT, HOT SPOT · RING OF FIRE

SETTING	MELTING MECHANISM	EXPRESSION	
Divergent	decompression	ridge, rift, basalt	
Subduction	slab volatile flux	trench-slab-arc system	
Plume	mantle upwelling	flood basalt, hot spot	
BENEFITS: new crust/land	soil parent	geothermal energy	minerals

CLASSIFICATION
- MELTING MECHANISM
- ridge, rift, basalt
- slab volatile flux
- trench-slab-arc system

DIAGNOSTIC BASIS
- mantle upwelling
- flood basalt, hot spot
- Ring of Fire = trench + descending slab + earthquakes + volcanic arc
- Plume model: broad head may form flood basalt; tail may form age chain.

PROCESS LINK
- Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.
- HAZARDS: lava
- density currents
- CLIMATE: stratospheric sulphate aerosol can cause temporary cooling

EXAM TRAP
- BENEFITS: new crust/land
- soil parent
- geothermal energy
- Impact converges through wind, relief, drainage, exposure and preparedness.

MECHANISM / VERDICT — Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

06

STAGE 06 INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK

INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK · INDIA REGIONAL HIERARCHY · COLLISION, THRUSTS, SLOPES, DENSE CORRIDORS · CONVERGENCE, STRIKE-SLIP COMPLEXITY · SUBDUCTION, TSUNAMI, BARREN ISLAND VOLCANISM · INHERITED RIFT AND FAULT REACTIVATION · RELATIVELY STABLE; KOYNA AND LATUR SHOW RESIDUAL RISK · ALLUVIUM, AMPLIFICATION AND HIGH EXPOSURE

INDIA REGIONAL HIERARCHY

Himalaya : collision, thrusts, slopes, dense corridors → North-East : convergence, strike-slip complexity → Andaman : subduction, tsunami, Barren Island volcanism

SYSTEM / DEFINITION
- INDIA REGIONAL HIERARCHY
- Himalaya : collision, thrusts, slopes, dense corridors
- North-East : convergence, strike-slip complexity
- Andaman : subduction, tsunami, Barren Island volcanism
- Kachchh : inherited rift and fault reactivation

PROCESS / MECHANISM
- Peninsula : relatively stable; Koyna and Latur show residual risk
- Indo-Gangetic : alluvium, amplification and high exposure
- MACROZONATION: broad design hazard, not prediction
- MICROZONATION: geology + soil + motion + water + liquefaction + GIS
- NCS/MoES → monitoring

SPATIAL / INDIA PATTERN
- GSI → geology
- BIS → standards
- NDMA/NIDM → guidance
- INCOIS → tsunami warning
- States/ULBs → approvals + inspection + enforcement site → regular load path

HAZARD / LIMIT / RESPONSE
- non-structural safety → quality control → retrofit → drills
- Early warning detects rupture after it starts; it is not prediction.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

06

STAGE 06 INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK

INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK | INDIA REGIONAL HIERARCHY | COLLISION, THRUSTS, SLOPES, DENSE CORRIDORS | CONVERGENCE, STRIKE-SLIP COMPLEXITY | SUBDUCTION, TSUNAMI, BARREN ISLAND VOLCANISM | INHERITED RIFT AND FAULT REACTIVATION | RELATIVELY STABLE; KOYNA AND LATUR SHOW RESIDUAL RISK | ALLUVIUM, AMPLIFICATION AND HIGH EXPOSURE

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PROCESS / MECHANISM

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- GSI → geology
- BIS → standards
- NDMA/NDM → guidance
- INCOIS → tsunami warning
- States/ULBs → approvals + inspection + enforcement site → regular load path → ductility → connections

HAZARD / LIMIT / RESPONSE

- non-structural safety → quality control → retrofit → drills
- Early warning detects rupture after it starts; it is not prediction.

MECHANISM / VERDICT — Macrozonation tells a city the regional design problem; microzonation tells neighbourhoods how local ground modifies that problem.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Macrozonation tells a city the regional design problem; microzonation tells neighbourhoods how local ground modifies that problem.

07

STAGE 07 HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA | WHY CAN SIMILAR EARTHQUAKES PRODUCE UNEQUAL DISASTERS? | SOURCE SIZE + DEPTH + RUPTURE GEOMETRY + PATH | BASIN SEDIMENT + GROUNDWATER + SLOPE + LIQUEFACTION POTENTIAL | PEOPLE + BUILDINGS + LIFELINES + TIME OF OCCURRENCE | WEAK DESIGN + IRREGULAR FORM + POOR MAINTENANCE | CODES + ENFORCEMENT + RETROFIT + WARNING + RESPONSE | RISK VERDICT

QUESTION: Why can similar earthquakes produce unequal disasters? → HAZARD: source size + depth + rupture geometry + path → SITE: basin sediment + groundwater + slope + liquefaction potential → EXPOSURE: people + buildings + lifelines + time of occurrence → VULNERABILITY: weak design + irregular form + poor maintenance

SYSTEM / DEFINITION

- QUESTION: Why can similar earthquakes produce unequal disasters?
- HAZARD: source size + depth + rupture geometry + path
- SITE: basin sediment + groundwater + slope + liquefaction potential
- EXPOSURE: people + buildings + lifelines + time of occurrence

PROCESS / MECHANISM

- VULNERABILITY: weak design + irregular form + poor maintenance
- CAPACITY: codes + enforcement + retrofit + warning + response
- RISK VERDICT: hazard becomes disaster through social and spatial filters
- 15-MARK INDIA SPINE tectonic regions → site modifiers → exposure/vulnerability → cases

SPATIAL / INDIA PATTERN

- zonation and microzonation → construction and governance → conclusion
- TRAPS: focus is not epicentre
- magnitude is not intensity tsunami is not a tide
- caldera is not an ordinary crater zone map is not prediction

HAZARD / LIMIT / RESPONSE

- code publication is not enforcement.

MECHANISM / VERDICT — RISK VERDICT: hazard becomes disaster through social and spatial filters

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: zonation and microzonation → construction and governance → conclusion.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | ADVANCED 1 — FINITE-FAULT RUPTURE, DIRECTIVITY AND FREQUENCY-DEPENDENT SITE | ADVANCED 2 — SEISMIC CYCLE, PALAEOSEISMOLOGY AND PROBABILITY | ADVANCED 3 — CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE | ADVANCED 4 — EARTHQUAKE EARLY WARNING AND OPERATIONAL LIMITS | ADVANCED 5 — VOLCANIC MONITORING AND ERUPTION-STYLE UNCERTAINTY | DIRECTIVITY HELPS EXPLAIN ELONGATED DAMAGE PATTERNS THAT A SIMPLE | FREQUENCY CONTENT MATTERS

OPTIONAL PROCESS DEPTH

- ADVANCED 1 — FINITE-FAULT RUPTURE, DIRECTIVITY AND FREQUENCY-DEPENDENT SITE RESPONSE
- ADVANCED 2 — SEISMIC CYCLE, PALAEOSEISMOLOGY AND PROBABILITY
- ADVANCED 3 — CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE
- ADVANCED 4 — EARTHQUAKE EARLY WARNING AND OPERATIONAL LIMITS

DATA / INSTITUTION LIMIT

- ADVANCED 5 — VOLCANIC MONITORING AND ERUPTION-STYLE UNCERTAINTY
- Directivity helps explain elongated damage patterns that a simple epicentral circle cannot.
- Frequency content matters: short stiff structures respond differently from tall flexible structures.
- Resonance occurs when dominant ground periods approach a structure's natural period.

CONTEMPORARY PARALLEL

- Basin-edge effects can focus or convert waves; deep basins may prolong long-period shaking.
- Nonlinearity: very strong shaking can change soil stiffness and damping, so weak-motion amplification cannot always be extrapolated directly.
- earthquake swarms and volcanic tremor;
- deformation from GPS, tilt and satellite radar;

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels